STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 6: System of Particles and Rotational Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Define centre of mass. [1]
Torque magnitude? [1]
SI unit of moment of inertia? [1]
Rotational kinetic energy? [1]
Condition for angular momentum conservation? [1]
Rolling relation? [1]
Static rigid-body equilibrium? [1]
Parallel-axis theorem? [1]

2 and 3 kg at x=0 and 5 m: x_cm? [2]
4 N perpendicular at .5 m: torque? [2]
I=3 kg m^2, omega=4: rotational energy? [2]
Radius .4 m wheel at 8 m/s: omega? [2]
Torque 12 N m and I=4: ? [2]
No external force: total momentum? [2]

2 kg at x=0 and 6 kg at x=4: x_cm? [3]
.25 m spanner, 40 N at 30 deg : torque? [3]
Solid disc M=4 kg, R=.5 m: I and KE at omega=10? [3]
omega rises 2 to 10 rad/s in 4 s: and theta? [3]

2 kg at x=0 and 3 kg at x=5 m, external force 10 N: x_cm and acceleration? [4]
Skater I changes 4 to 1.6 kg m^2 from omega=3: final omega and KE change? [4]

End of paper - check working, units, signs and significant figures.